

Shelf Scan — Product Requirements Document

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1. Overview - The Problem

Cataloguing a bookshelf in a bookstore during inventory, a friend's home library, or one's own collection after a move - remains a manual, repetitive task. Each spine must be read individually: titles running vertically, lettering faded or worn, authors that require pulling the book out to confirm. This repeats for every title on the shelf, whether the collection holds thirty books or two hundred, with no available shortcut.

Shelf Scan replaces this manual process with a single photograph. Pointing a camera at a shelf returns a structured catalogue automatically: title, author, genre, and - where available - publisher, ISBN, page count, and a summary, with any uncertain read flagged for review rather than guessed silently.

2. Goals

- Primary: identify every visible book on a shelf from one photo, give title, author, genre, confidence score at an accuracy where manual correction is the exception, not the norm.
- Secondary: enrich each identified book with catalogue enrichment (ISBN, publisher, page count etc) without requiring any extra user effort.
- Reliability: fail honestly. Every error state should tell the user what actually happened, not a plausible-sounding guess.

3. Success Metrics / Acceptance Criteria

These are the targets this build was designed and tested against:

- 100% of returned output carries an explicit confidence label (high / medium / low) - never an unqualified guess.
- 0% of clearly non-bookshelf photos (furniture, people, outdoor scenes) are silently processed as "0 books found" - each is explicitly classified and explained with an error message.
- A response that is cut off mid-scan (large shelf, long titles) still returns every complete book read before the cutoff, with the user told the result may be incomplete - never a hard failure that discards a partially successful scan.
- A failed catalogue-enrichment lookup (extra data) never blocks or breaks the core title/author/genre/confidence result.

Note: these are engineering acceptance criteria validated through structured manual testing during the build (including deliberately provoking failure cases), not a large-scale statistical benchmark - see Section 9 for what a proper accuracy benchmark would require.

4. Users & Use Cases

- Personal library owners cataloguing a home collection
- Someone moving house who needs a fast inventory of books
- Small independent or used bookstores doing quick stock intake

5. Scope

In scope (v1)

- Camera capture (mobile) and gallery upload, with in-browser cropping before scan
- AI-based spine reading: title, author, genre, per-book confidence
- Scene classification: bookshelf / wrong-shelf-type / not-a-shelf, each with a tailored response
- Human-in-the-loop correction: any field is editable; saving a correction sets confidence to high and re-triggers enrichment
- Catalogue enrichment via Open Library: publisher, ISBN, page count, edition count, cover art, summary (built; currently environment-gated - see Section 8)

Out of scope (v1)

- Live ratings : neither Amazon nor Goodreads offer a free public API

- Barcode/ISBN scanning of the physical copy : matching is by title/author, so edition data is approximate, not tied to the exact physical copy
- User accounts or saved catalogues (session-only)
- Standalone hosting outside Claude (requires a backend proxy)

6. Functional Requirements

6.1 Capture

- Users can capture a photo directly from the device camera or select an existing photo from the gallery.
- Users can crop the photo before submitting. This is an accuracy lever, not just UX polish - a tighter crop measurably improves spine legibility for the model.

6.2 Classify

- The photo is classified before any spine-reading is attempted: bookshelf (has books), other_shelf (shelving without books - e.g. clothes, drinks), or not_shelf (unrelated image).
- other_shelf and not_shelf responses describe what was actually seen, rather than a generic “try again.”

6.3 Read

- For each identified book: title, author, genre, and a confidence level reflecting spine legibility, not correctness.
- Output parsing must tolerate model response variability (stray commentary, markdown fences, truncation) and degrade gracefully rather than fail outright.

6.4 Enrich

- Each book is looked up by title + author against Open Library for publisher, ISBN, page count, edition count, cover image, and summary.
- Enrichment failures are silent to the user experience — the core read always renders regardless of lookup success.

6.5 Review

- Any field is user-editable per book. Saving upgrades that book to high confidence and re-runs enrichment against the corrected title/author.

7. Error Handling

Each of these came from a real failure encountered during testing, not a hypothetical:

- Rate-limit (429) responses must be distinguished from generic failures and explained accurately - early builds misattributed this to “too many books” and to “connection issues” respectively; both were wrong diagnoses that cost debugging time.
- A truncated response and a genuine parse failure must produce different messages, because they have different fixes (split the photo vs. simply retry).
- An unreachable third-party API (catalogue enrichment) must never be reported as “no match found” - that conflates “couldn't check” with “checked and it doesn't exist,” which is a meaningfully different, misleading claim.

8. Technical Constraints & Trade-offs

Shelf Scan runs as a Claude AI-powered artifact: the browser calls Claude's vision API directly, authenticated through the viewer's own Claude session, so no API key ever ships in client-side code. The trade-off is that this restricts the app to Claude's hosting environment, which only permits calls to Claude's own API - which is why the fully-built Open Library enrichment doesn't yet execute live. Removing this constraint requires a small backend proxy holding a real Anthropic API key, which would also simultaneously unlock the enrichment calls, since the app would no longer be sandboxed.

9. Risks & Open Questions

- Open Library coverage is thinner for self-published and regional titles than for widely-published ones.
- Very large shelves (25+ books in one frame) may need to be split across two photos to stay within a single response's length budget.
- Accuracy is currently validated by manual spot-checking against real shelves, not a labeled benchmark set - a proper accuracy number requires building a small “Golden Dataset” of photographed shelves with verified ground truth, the same evaluation approach I used professionally for Menu Recognition. This is a WIP I am doing for ShelfScan.

10. Roadmap

- Backend proxy for standalone hosting, unlocking Open Library enrichment
- Google Books API as a secondary source for ratings
- Export catalogue as CSV / JSON
- A labeled benchmark set to move accuracy from spot-checked to measured